D02 · Inventory Strategy & Optimization

01 · ABC-XYZ segmentation & inventory policy design

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

ABC-XYZ segmentation and policy design classify items by both value (ABC) and demand variability (XYZ), then assign each segment a differentiated inventory policy — service level, review mode, and safety-stock formula — so scarce working capital is positioned where it buys the most service, instead of a single blanket policy applied to a non-uniform catalog.

2 WHY IT MATTERS

Arrow, Harris & Marschak 1951 **(Arrow, Harris & Marschak, 1951 — peer-reviewed)** established the foundations of optimal inventory policy under uncertainty, showing that the right safety stock is a function of demand variability and the cost of a stockout relative to holding — not a flat target. A single service level across the catalog ignores this: it over-protects high-value predictable items (whose low variability needs little buffer) and under-protects erratic items (whose high variability needs more), inverting where the capital should sit. The cost is silent and large — **a uniform 95% service target typically over-invests in the A/X corner and starves the C/Z corner**, paying for inventory that buys no incremental service while missing on the items that actually stock out.

3 HOW IT IS DONE

1. **Segment on two axes, not one.** ABC ranks items by value or volume; XYZ ranks them by demand variability (coefficient of variation). Value alone tells you how much attention an item deserves; variability tells you how much buffer it needs. The policy lives in the 3×3 grid, not in ABC alone.
2. **Differentiate the service level by segment.** Set higher service targets where a stockout is costly and demand is predictable enough to protect efficiently, and lower targets where buffering is expensive relative to the value. A single service level is the most expensive way to set inventory.
3. **Match the safety-stock formula to the demand pattern.** Smooth high-volume items use a standard normal-based safety stock; intermittent items need a method that handles their distribution (the same reasoning that drives Croston-class forecasting). The wrong formula mis-sizes the buffer regardless of the service target.
4. **Choose the review/replenishment mode per segment.** Continuous review (s,Q) for high-value items where monitoring pays; periodic review (R,S) for low-value items where review cost dominates. The control mode is part of the policy, not a system default.
5. **Re-segment on a cadence.** Items migrate across the grid as they age — a new A/Y becomes a C/Z near end-of-life. A segmentation frozen at implementation positions capital for a catalog that no longer exists; tie re-segmentation to the planning cycle.
6. **Validate the service-cost trade-off against actuals.** Check realized service and inventory by segment against the policy intent. If the A/X corner runs at 99% while C/Z misses, the policy is mis-positioned — measure the trade-off, do not assume the formula delivered it.

> **WORKED EXAMPLE — ILLUSTRATIVE**
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> A distributor ran a flat 95% service target across 9,000 SKUs — **baseline: $41M inventory, 95% aggregate fill, but A/X items at 99% and C/Z items at 88%**. It built the ABC-XYZ grid, raised service on cost-effective segments, lowered it on expensive-to-buffer C/Z, and matched safety-stock formulas to pattern. Results: **inventory $41M -> $34M (-17%), aggregate fill 95% -> 96%, and the service distribution flattened** — the same service bought with less capital, because the capital finally sat where it earned service.

IN PRACTICE

A policy map for the grid

Define, for each of the nine cells, the service target, review mode, and safety-stock method. The map turns segmentation from an analysis into an operating policy — every item inherits a deliberate treatment rather than a system default carried over from go-live.

Scheduled re-segmentation

Re-run the segmentation on a regular cadence and reassign migrated items. Lifecycle movement is constant; a static segmentation slowly positions capital for the catalog of two years ago. Tie re-segmentation to the same cadence as forecast re-classification.

Service-by-segment monitoring

Report realized service and inventory by segment, not just in aggregate. Aggregate service hides the inversion — a healthy average can mask over-served A-items and under-served C-items. Segment-level reporting is the only view that reveals whether the policy is doing what it intended.

Align segmentation with forecasting and MEIO

The variability axis that drives policy is the same one that drives forecast model choice, and the segment policy is the input to multi-echelon optimization. Keep one consistent segmentation across forecasting, policy, and network optimization so the three are not silently working from different classifications.

WHAT YOU WOULD USE

INVENTORY POLICY / OPTIMIZATION

SAP IBP Inventory / o9 / Kinaxis — enterprise tier-1 — Native segmentation, differentiated service-level policy, and optimization-based safety stock integrated with the planning model. **(SAP, o9, Kinaxis — vendor)** The choice when inventory policy must feed the same engine that runs demand and supply.

Blue Yonder / RELEX — inventory optimization — Strong segmentation and inventory optimization, with retail-grade store/DC policy in RELEX. **(Blue Yonder, RELEX — vendor)** A strong fit where item-location policy granularity and replenishment are the priority.

Inventory-optimization specialists — Dedicated inventory-optimization tools focused on segmentation, safety-stock math, and policy tuning. **(specialized vendors — vendor)** The vertical option when the core APS lacks rigorous, pattern-aware safety-stock methods.

For operations that don't need a platform — An ABC-XYZ grid and a documented policy map per cell, with safety stock computed per pattern in a workbook and reviewed each cycle. The differentiation logic — not the optimizer — captures most of the working-capital gain; the math is well understood and runs fine in a spreadsheet for a bounded catalog.

4 SO WHAT

Operations that set policy on the value-and-variability grid buy a given service level with less capital; operations that run one blanket service target over-invest in the predictable corner and stock out in the erratic one. The flat service target feels fair and is quietly the most expensive policy available — it pays for buffer where none is needed and withholds it where it is. Segmentation is not analytics housekeeping; it is where the working-capital efficiency is.

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